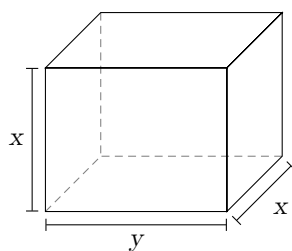


Problem 4

Problem 4

Problem 1 A box with two square sides is constructed to have a volume of 27. Denote the dimensions of the length of the base as y , and the width and height as x , as labeled in the diagram below. Find the dimensions needed to minimize the surface area of the box.



Solve the problem by performing the following steps.

- Find a formula for $S(x)$ the surface area of the box, as a function of only the variable x .
- Find the interval of interest for $S(x)$.
- Find the x -value that gives the minimum surface area over all such boxes.

[2]